

Large Language Models (LLM) vs Small Language Models (SLM) – Summary Paper

Introduction

Large Language Models (LLMs) and Small Language Models (SLMs) are AI models designed to understand and generate human language. They differ mainly in model size, computing requirements, capabilities, and deployment options.

Large Language Models (LLMs)

LLMs are trained on massive datasets and contain billions or more parameters. Examples include GPT models, Claude, Gemini, and large open-source models.

Advantages: Strong reasoning, broad knowledge, advanced text generation, coding support, summarisation, and multi-task capability.

Limitations: Higher training cost, greater hardware requirements, increased latency, and more complex deployment.

Small Language Models (SLMs)

SLMs are compact models designed for efficiency and specialised applications. They require fewer computing resources and can often run on local or edge devices.

Advantages: Lower cost, faster responses, improved privacy, easier deployment, and suitability for domain-specific applications.

Limitations: Lower general intelligence and reasoning capability compared with large models.

Key Differences

Model Size: LLMs have billions of parameters, while SLMs have fewer parameters.

Performance: LLMs generally provide stronger reasoning and broader knowledge.

Cost: SLMs are more economical to train and operate.

Deployment: LLMs commonly run in cloud environments; SLMs can run on local systems and edge devices.

Applications: LLMs support general AI assistants, while SLMs are ideal for focused enterprise tasks.

Recommendation

Use LLMs when advanced reasoning, flexibility, and broad knowledge are required. Use SLMs when cost efficiency, privacy, low latency, and specialised performance are priorities. A hybrid approach using LLMs for complex tasks and SLMs for targeted workloads is recommended for many enterprises.

Conclusion

LLMs deliver maximum capability and versatility, while SLMs provide efficiency and easier deployment. The right choice depends on business goals, data requirements, budget, and operational constraints.